List

In [26]:
```
l1 = []
l1
```

Out[26]:
```
[]
```

In [28]:
```
print(type(l1))
```

```
<class 'list'>
```

In [30]:
```
l2 = [20, 45, 30]
```

In [32]:
```
l3 = [13.44, 56.44, 56.11]
```

In [34]:
```
l4 = ['one', 'two', "three"]
l5 = ['arya', 26,[23,45],[120,78]]
l6 = [122, 'arya', 23.444]
l7 = ['arya', 26,[23,45],[120,78] , {'john', 'david'}]
```

In [36]:
```
len(l6)
```

Out[36]:
```
3
```

In [38]:
```
len(l3)
```

Out[38]:
```
3
```

In [42]:
```
l2[0]
```

Out[42]:
```
20
```

In [44]:
```
l4[0]
```

Out[44]:
```
'one'
```

In [46]:
```
l4[0][0]   # nested indexing-- access the first character of the first list elem
```

Out[46]:
```
'o'
```

In [48]:
```
l4[-1]
```

Out[48]:
```
'three'
```

In [52]:
```
l5[-1]
```

Out[52]:
```
[120, 78]
```

list slicing

In [151…
```
list = ['shraddha', 'pranali', 'gaytri', 'snehal', 'nirjala', 'vaishnavi', 'rohi
```

In [153…
```
list[0:3]
```
Out[153…
```
['shraddha', 'pranali', 'gaytri']
```

In [155…
```
list[2:5]
```
Out[155…
```
['gaytri', 'snehal', 'nirjala']
```

In [157…
```
list[:3]
```
Out[157…
```
['shraddha', 'pranali', 'gaytri']
```

In [159…
```
list[:2]
```
Out[159…
```
['shraddha', 'pranali']
```

In [161…
```
list[-3:]
```
Out[161…
```
['rohini', 'kaveri', 'anuja']
```

In [163…
```
list[-2:]
```
Out[163…
```
['kaveri', 'anuja']
```

In [165…
```
list[-1]
```
Out[165…
```
'anuja'
```

In [167…
```
list[:]
```
Out[167…
```
['shraddha',
 'pranali',
 'gaytri',
 'snehal',
 'nirjala',
 'vaishnavi',
 'rohini',
 'kaveri',
 'anuja']
```

Add, Remove and Change Items

In [170…
```
list
```
Out[170…
```
['shraddha',
 'pranali',
 'gaytri',
 'snehal',
 'nirjala',
 'vaishnavi',
 'rohini',
 'kaveri',
 'anuja']
```

In [172…
```
list.append('shreya')   # add on item to existing list
list
```

Out[172…
```
['shraddha',
 'pranali',
 'gaytri',
 'snehal',
 'nirjala',
 'vaishnavi',
 'rohini',
 'kaveri',
 'anuja',
 'shreya']
```

In [174…
```
list.insert(10,'nikita')  #add item at index location 10
list
```

Out[174…
```
['shraddha',
 'pranali',
 'gaytri',
 'snehal',
 'nirjala',
 'vaishnavi',
 'rohini',
 'kaveri',
 'anuja',
 'shreya',
 'nikita']
```

In [176…
```
list.insert(6,'sidhi')  #add item at index location 6 and move existing item to
list
```

Out[176…
```
['shraddha',
 'pranali',
 'gaytri',
 'snehal',
 'nirjala',
 'vaishnavi',
 'sidhi',
 'rohini',
 'kaveri',
 'anuja',
 'shreya',
 'nikita']
```

In [178…
```
list.remove('sidhi')   # remove is function method we can not use square bracket
list     # it will remove item sidhi
```

Out[178…
```
['shraddha',
 'pranali',
 'gaytri',
 'snehal',
 'nirjala',
 'vaishnavi',
 'rohini',
 'kaveri',
 'anuja',
 'shreya',
 'nikita']
```

In [180…
```
list.pop()   # pop---remove last item of list
list
```

Out[180…
```
['shraddha',
 'pranali',
 'gaytri',
 'snehal',
 'nirjala',
 'vaishnavi',
 'rohini',
 'kaveri',
 'anuja',
 'shreya']
```

In [182…
```
list.pop(9)   # it remove last item shreya
list
```

Out[182…
```
['shraddha',
 'pranali',
 'gaytri',
 'snehal',
 'nirjala',
 'vaishnavi',
 'rohini',
 'kaveri',
 'anuja']
```

In [184…
```
del list[7]    # removes item at index 7
list
```

Out[184…
```
['shraddha',
 'pranali',
 'gaytri',
 'snehal',
 'nirjala',
 'vaishnavi',
 'rohini',
 'anuja']
```

In [186…
```
# change value of the string
list[0] = 1
list[1] = 2
list[2] = 3
list
```

Out[186…
```
[1, 2, 3, 'snehal', 'nirjala', 'vaishnavi', 'rohini', 'anuja']
```

In [188…
```
list.clear()   # delete all items in the list
list
```

Out[188…
```
[]
```

In [192…
```
del list
list
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[192], line 1
----> 1 del list
      2 list

NameError: name 'list' is not defined
```

Copy List

In [195…
```
list = ['shraddha', 'pranali', 'gaytri', 'snehal', 'nirjala', 'vaishnavi', 'rohi
```

In [197…
```
list1 = list
```

In [199…
```
id(list), id(list1)   # when we show two list equal or same the address of both
```

Out[199…
```
(2559558120832, 2559558120832)
```

In [201…
```
list[0]
```

Out[201…
```
'shraddha'
```

In [203…
```
list
```

Out[203…
```
['shraddha',
 'pranali',
 'gaytri',
 'snehal',
 'nirjala',
 'vaishnavi',
 'rohini',
 'kaveri',
 'anuja']
```

In [207…
```
list1
```

Out[207…
```
['shraddha',
 'pranali',
 'gaytri',
 'snehal',
 'nirjala',
 'vaishnavi',
 'rohini',
 'kaveri',
 'anuja']
```

Join Lists

In [210…
```
l1 = ['shraddha','priya','gaytri','pranali']
l2 = ['renuka','snehal','nirjala','nikita']
```

In [212…
```
l3 = l1 + l2    # join --- it will join two list using + operator
l3
```

Out[212…
```
['shraddha',
 'priya',
 'gaytri',
 'pranali',
 'renuka',
 'snehal',
 'nirjala',
 'nikita']
```

In [214…
```
l1.extend(l2)    # append l2 with l1
l1
```

Out[214…
```
['shraddha',
 'priya',
 'gaytri',
 'pranali',
 'renuka',
 'snehal',
 'nirjala',
 'nikita']
```

List Membership

In [217…
```
l1
```

Out[217…
```
['shraddha',
 'priya',
 'gaytri',
 'pranali',
 'renuka',
 'snehal',
 'nirjala',
 'nikita']
```

In [219…
```
'shraddha' in l1   # it will check is 'shraddha' is existing in l1
```

Out[219…
```
True
```

In [221…
```
'ten' in l1    # ten not exist in l1 so it will give false o/p
```

Out[221…
```
False
```

In [223…
```
if 'three' in l1:
    print('three is present in list')
else:
    print('three is not present in list')
```

```
three is not present in list
```

In [225…
```
if 'three' in l1:
    print('three is present in list')
else:
    print('three is not present in list')
```

```
three is not present in list
```

Reverse and Sort List

In [228…
```
l1
```

Out[228…
```
['shraddha',
 'priya',
 'gaytri',
 'pranali',
 'renuka',
 'snehal',
 'nirjala',
 'nikita']
```

In [230…
```
l1.reverse()   # it will print reverse string 1-10 to 10-1
l1
```

Out[230…
```
['nikita',
 'nirjala',
 'snehal',
 'renuka',
 'pranali',
 'gaytri',
 'priya',
 'shraddha']
```

In [232…
```
l1 = l1[::-1]  # reverse the list
l1
```

Out[232…
```
['shraddha',
 'priya',
 'gaytri',
 'pranali',
 'renuka',
 'snehal',
 'nirjala',
 'nikita']
```

In [234…
```
l3 = [7,89,67,4,0,6,12]
l3.sort()   # sort-- it will list items of list in ascending order
l3
```

Out[234…
```
[0, 4, 6, 7, 12, 67, 89]
```

In [236…
```
l3 = [7,89,67,4,0,6,12]
l3.sort(reverse=True)   # sort-- it will list items of list in descending order
l3
```

Out[236…
```
[89, 67, 12, 7, 6, 4, 0]
```

In [238…
```
l4 = [12,23,45,56,89,25]
sorted(l4)   # returns new sorted list and dosen't change orignal list
```

Out[238…
```
[12, 23, 25, 45, 56, 89]
```

In [240…
```
l4
```

Out[240…
```
[12, 23, 45, 56, 89, 25]
```

Loop through a list

In [243…
```
l1
```

Out[243…
```
['shraddha',
 'priya',
 'gaytri',
 'pranali',
 'renuka',
 'snehal',
 'nirjala',
 'nikita']
```

In [245…
```
for i in l1:
    print(i)
```

```
shraddha
priya
gaytri
pranali
renuka
snehal
nirjala
nikita
```

In [247…
```
for i in enumerate(l1):   # enumerate--- it will print both index and values whi
    print(i)
```

```
(0, 'shraddha')
(1, 'priya')
(2, 'gaytri')
(3, 'pranali')
(4, 'renuka')
(5, 'snehal')
(6, 'nirjala')
(7, 'nikita')
```

Count

In [250…
```
l20 = [23,45,23,78,45,14,95,19]
```

In [252…
```
l20.count(23)  # count-- it will count how many thime the item apppears in list
```

Out[252…
```
2
```

In [254…
```
l20.count(19)
```

Out[254…
```
1
```

In [256…
```
l20.count(45)
```

Out[256…
```
2
```

All/Any

In [259…
```
l1 = [2,3,4,5]
```

In [261…
```
all(l1)
```

Out[261…
```
True
```

In [263…
```
l1 = [2,3,4,5,0]
```

In [265…
```
all(l1)  # return false if one value is false(value 0)
```

Out[265…
```
False
```

In [271…
```
any(l1)   # any--- return true if we have in the list with true values
```

Out[271…
```
True
```

In [267…
```
l2 = [1,2,3,True,False]
```

In [269…
```
all(l2)
```

Out[269…
```
False
```

In [273…
```
any(l2)
```

Out[273…
```
True
```

In [ ]: